PCS-9567MV-10000

Medium Voltage Skid

Operation and Maintenance Manual

NR Electric Co., Ltd.

Preface

About This Manual

The manual describes the PCS-9567MV-10000 Medium Voltage (MV) Skid in terms of safety information, product introduction, operation and maintenance, etc. Please read all safety information and get familiar with the functions and features before proceeding with the operations of this product.

Intended Audience

This manual is intended for the persons who are required to install, configure or service equipment, or any other associated operation, it assumes a reasonable level of understanding in these disciplines.

Only qualified personnel are allowed to install, operate and maintain this equipment. Qualified personnel are individuals who:

- Are familiar with the installation, commissioning, and operation of the equipment and of the system to which it is being connected.
- Are able to safely perform switching operations in accordance with accepted safety engineering practices and are authorized to energize and de-energize equipment and to isolate, ground, and label it.
- Are trained in the care and use of safety apparatus in accordance with safety engineering practices.
- Are trained in emergency procedures (first aid).

Symbols

Safety information is highlighted with the following symbols according to the degree of danger:

⚠ DANGER	Indicates an imminently hazardous situation that, if not avoided, will result in death or serious injury.
⚠ WARNING	Indicates a potentially hazardous situation that, if not avoided, could result in death or serious injury.
⚠ CAUTION	Indicates a potentially hazardous situation that, if not avoided, may result in minor or moderate injury or equipment damage.

NOTICE	Indicates that property damage can result if the measures specified are not taken.
INFORMATION	Important information about the product, please pay attention to avoid undesired result.

Document Revision History

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R1.00	2025-03-11	First release

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1 Safety Instruction

This chapter presents the safety instructions for PCS-9567MV-10000 medium voltage skid (abbreviated as **"MV skid"**, or **"this product"**). Read this document thoroughly before operations. NR will not be liable for any damage caused by violation of the safety instructions in this document.

1.1 Personnel Requirements

- Only certified and professionally authorized personnel are allowed to work on or near the MV skid. All installation and service personnel must be trained to be familiar with the precautions and workflows mentioned in this manual, as well as the safety regulations.
- Strictly abide by all instructions contained in this manual, as well as the rules and laws of the country/region where the project is located.
- The minimum required safety equipment is as follows: safety shoes, safety gloves, safety glasses, head protection, work clothes. Take special precautions when you work close to exposed conductors.
- Only personnel with corresponding qualifications are permitted for special operations. Necessary safety precautions (such as fastening safety belts for climbing operations) must be taken.
- In case of abnormal conditions (such as abnormal noise, sudden bad weather, etc.), the operation should be stopped.
- Keep all doors locked while this product is operating. Spend as little time as possible near this product.

1.2 Electrical Connection

- Power cables and communication cables should be placed in different cable trenches. Communication cables should be laid as far away as possible from the power cables, and avoid power cables and communication cables walking parallel to each other for a long distance, so as to reduce electromagnetic interference caused by output voltage transients.
- There is large current in the power cable, so the damage of the cable insulation should be avoided during the installation process, so as to avoid short circuit. Moreover, the power cable should be fixed properly, and the connected cable should be carefully handled to avoid exerting excessive external force.
- Do NOT touch exposed terminals when the MV skid is energized, so as to avoid possible dangerous high voltage.

- Dangerous voltages may still be present in the DC circuit after the power is turned off in this product. These voltages will not disappear within several seconds.
- When connecting the AC voltage/current circuit or DC power supply circuit, please confirm the consistency with the nominal parameters of this product.
- When this control device is energized, it is forbidden to insert or unplug the PCB (Printed Circuit Board), otherwise it may cause the device to operate incorrectly.
- When connecting the output contacts of this control device to external circuit, the external power supply voltage should be carefully checked, so as to prevent the connected circuit from overheating.
- The grounding cable must be grounded reliably! Otherwise, this product may fail to work, and even may cause fatal shock hazard to the operator in case of malfunction.

1.3 Daily Operations

⚠ DANGER

The DC voltage may reach 1500V and the AC voltage may reach several tens of kilovolts when this product is running, and it will be dangerous to touch the live part of this product directly. Improper operation may lead to serious personal injury.

⚠ WARNING

Do NOT modify parameters of the MV skid without the permission of NR. Unauthorized modification of functional parameters may cause personal injury or equipment damage, in which case NR will not provide warranty service.

⚠ WARNING

When the primary system is in live operation, secondary side open circuit of the current transformer (CT) connected to this product is absolutely forbidden. Open circuit may cause extremely dangerous high voltages.

1.4 Maintenance & Replacement

- All maintenance and repair operations can only be performed on the equipment when the MV skid is completely disconnected from the battery system and the power grid, and it is confirmed that these power supplies will not be turned on again and the capacitor discharging is completed.

- Energy storage battery side open-circuit voltage must NOT exceed the maximum permissible input voltage, otherwise this product may be damaged.
- Grid-connected charging and discharging of MV skid should be approved by the local power supply department, and the relevant operation must be performed by certified and professionally authorized personnel.
- During the normal operation of the MV skid, do NOT disconnect the AC side circuit breaker (abbreviated as CB) or the DC side switch directly at will, otherwise personal injury or equipment damage may occur!
- Do NOT cut off the power supply of the PCS (Power Conversion System) control unit during operation. If it is required to cut off the power, please follow the shutdown operation process.
- Do NOT let the PCS run in the overload situation for a long time, please pay attention to the temperatures of the PCS and power modules in case of overload. If the temperature of the power module is high, please reduce the P (active power) value of the PCS remote regulation, so as to ensure the safety of operation.
- During maintenance, please make sure that the MV skid will not be energized accidentally. It is recommended to take insulation protection measures when contacting possible live parts.
- Clean the air filters in time to ensure smooth air flow.

1.5 Environmental Requirements

- The installation environment is outdoor.
- Please make sure that the installation environment is free from shaking and vibration.
- The foundation must meet the design requirements, ensuring it is solid, safe and reliable. The foundation must have the bearing capacity to support the weight of the MV skid. No depressions or inclinations are allowed on the ground surface. To prevent the risk of corrosion, the MV skid should be installed at a location higher than the surrounding environment, so that surface water does not accumulate around it.
- Adequate distance must be maintained between the installation location and walls or other equipment to meet the requirements for the maintenance access, escape route and ventilation.
- The installation environment should be as far away as possible from areas where people lives, as the MV skid may generate some noise during operation.
- Ensure that all external cable entries are completely sealed to prevent the ingress of foreign objects such as rats, insects and dust.
- During storage and installation, avoid unnecessarily opening the doors to protect the equipment from rain and dust.

2 Product Introduction

2.1 Overview

2.1.1 Technical Data

Transformer	
Rated Power (kVA)	10000• / 9000◦/ 8000◦
Transformer Model	Oil type[1]
Transformer Vector	Dy11-y11
Protection Level	IP54• / IP55◦
Anti-corrosion Grade	C4-H• [2] / C4-VH◦ / C5-M◦ / C5-H◦ / C5-VH◦
Cooling Method	ONAN• / ONAF◦
Temperature Rise	60K（Top Oil） 65K（Winding）@40℃
Oil Retention Tank	None• / Galvanized steel◦ [3]
Winding Material	Aluminum• / Copper◦
Transformer Oil	25# /45# mineral oil• / Natural ester insulation oil◦
Transformer Efficiency	IEC standard• / IEC Tier-2◦
MV Operating Voltage Range (kV)	11~33±5%
Nominal Frequency (Hz)	50 / 60
Altitude (m)	＜1000• / ＞1000◦
Switchgear	
Switchgear Type	Ring Main Unit, CCV [4]
Rated voltage (kV)	12/24/36
Insulating medium	SF6
Rated frequency (Hz)	50/60
Enclosure protection degree	IP3X
Gas tank protection degree	IP67
Gas leakage rate per year	≤0.1%
Rated Operating Current (A)	630
Switchgear Short Circuit Rating (kA/s)	20kA/3s• / 25kA/3s◦
Switchgear IAC (kA/s)	A FL 20kA 1S
PCS * 4	
DC Input Voltage Range (V)	1050～1500
Max. DC input Current (A)	1310*2
DC Voltage Ripple	< 2%
DC Current Ripple	< 3%
LV Nominal Operating Voltage (V)	690

LV Operating Voltage Range (V)	621～759
PCS Efficiency	98.7%[5]
Max. AC Output Current (A)	1151*2
Total Harmonic Distortion Rate	< 3%
Reactive Power Compensation	Four quadrant operation
Nominal Output Power (kVA)	1250*2
Max. AC Power (kVA)	1375*2
Power Factor Range	>0.99
Nominal Frequency (Hz)	50 / 60 Hz
Operating Frequency (Hz)	45~55 / 55~65 Hz
Connection Phases	Three-phase-three-wire
Protection	
DC Input Protection	Disconnector + Fuse inside of inverter
AC Output Protection	Motorized Circuit breaker inside of Inverter
DC Overvoltage Protection	Surge arrester, type II• / I+II◦
AC Overvoltage Protection	Surge arrester, type II• / I+II◦
Ground Fault Protection	DC IMD• / DC IMD+ AC IMD◦
Transformer Protection	Protection relay for pressure, temperature, gassing
Fire Extinguishing System	Smoke detector sensor (dry contact)
Communication Interface	
Communication Method	CAN / RS485 / RJ45 / Optical fiber
Supported Protocol	CAN / Modbus / IEC60870-103 / IEC61850
Ethernet Switch Qty	One for standard [6]
UPS	1kVA for 15min• / 1h◦ / 2h◦
Skid General	
Dimensions (W*H*D)(mm)	12192*2896*2438 (40ft)
Weight (kg)	38800
Protection Level	IP54
Operating Temperature (℃)	-35~60, >45 derating
Storage Temperature (℃)	-40~70
Maximum Altitude (above sea level) (m)	5000, ≥3000 derating [7]
Environment Humidity	0~ 100%, No condensation
Type of Ventilation	Nature air cooling• / Forced air cooling◦
Auxiliary Power Consumption (kVA)	21 (peak)
Auxiliary Transformer (kVA)	Without• / With◦[8]

INFORMATION

● **Standard** ○ **Optional**

[1] If dry transformer is required, please contact with NR for more information.

[2] Lower protection level will be covered by C4-H.

[3] Standard for no supply of oil retention tank. If required to be integrated with PCS skid, please contact with NR.

[4] If other type of switchgear is required, please contact with NR for more information. In **"CCV"**, **"C"** is the abbreviation for **"Cable load break switch unit"**, **"V"** is the abbreviation for **"Vacuum circuit breaker unit"**.

[5] Typical discharge value for each PCS running at DC 1200V under IEC 62933-2-1 environment condition.

[6] If more Ethernet switch is required, please contact with NR for more information.

[7] When altitude is between 3000~4000m, the system LV AC voltage shall be less than 600V; When altitude is between 4000~4500m, the system LV AC voltage shall be less than 550V; When altitude is between 4500~6000m, please contact with NR for more information.

[8] Please contact with NR for more information.

2.1.2 Functions

As a flexible interface between energy storage units and the power grid, the power conversion system (PCS) is developed using highly reliable and intelligent power modules. Through its integrated charging and discharging design, it enables bidirectional energy flow between AC and DC systems. NR PCS-9567 power conversion system can achieve the following functions:

- **Accurate and flexible charging & discharging control modes**

  Provides CAN, Ethernet and RS-485 interface for real-time communication with battery management system (BMS). It can accurately monitor current battery operation information, not only control the charging and discharging status of converter, but also switch to constant current mode, constant voltage mode, constant power mode and other charging & discharging modes conveniently, so as to achieve the optimal strategy for the battery charge and discharge control. Various types of energy storage batteries are supported: lithium-ion battery, sodium-ion battery, lead-acid battery, lead-carbon battery and etc.

- **Free switchover between grid-connected operation mode and grid-forming operation mode**

  Realizes bi-directional energy exchange with the power grid in grid-connected mode, also can support regional power grid in grid-forming mode. Moreover, it can be used as the main power supply to support the isolated grid. The 2 operation modes cab be switched freely.

- **Soft grid-connected control and power quality control**

  The control system provides accurate real-time control for the output voltage of the converter according to the monitored online power grid voltage. It can eliminate the static & dynamic errors, and realize the impact-free grid-connection. The optimized filter and harmonic suppression algorithm can realize the optimal control of grid-connected power and ensure the power quality.

- **Respond to EMS instructions for load leveling**

  Under the dispatching of EMS (Energy Management System), it can store electric energy during load valley and release electric energy during load peak, so as to reduce peak-valley difference of power grid, improve load characteristics of power grid, and realize power system load control and load shifting.

- **Power grid frequency and reactive power regulation**

  During grid-connected operation, it provides primary and secondary frequency regulation in coordination with AGC (Automatic Generation Control) or PMS (Power Management System). Moreover, it can realize steady-state reactive power control in coordination with AVC (Automatic Voltage Control), realize transient voltage control in coordination with PMS.

- **Complete self-supervision and protection functions**

  The scope of self-supervision covers the control system, I/O units, converter power modules, buses and other related equipment. The system fault can be detected within 1ms and corresponding blocking pulse or tripping signal will be issued if fault is detected. The complete protections ensure the normal operation of the converter.

- **Powerful communication function**

  The device is equipped with multiple optical interfaces and RJ45 interfaces, and supports high-speed communication protocols (e.g.: GOOSE and IEC 60044-8). The communication time delay is less than 2ms, and it can meet the fast control response requirement of some applications (e.g.: frequency regulation, voltage regulation, fluctuation smoothing and micro-grid control).

- **Transient fault recording function**

  The system can continuously record the signals before and after the fault. The recorded data files are stored in the shared directory of the operator workstation and can be used for fault analysis. The file format adopts the standard COMTRADE format.

2.1.3 Features

- It is developed based on high-performance high-reliability UAPC platform of NR and provides a friendly human-machine interface.

- Comprehensive charge/discharge limitation functions that coordinate with BMS to ensure battery safety during the charging and discharging process.
- Complete and reliable protection functions to guarantee the reliable and safe operation of the product, or to provide safe and prompt protective shutdown.
- High power quality, with grid-connected power quality far superior to the technical requirements specified in relevant standards.
- Adaptive to high altitude applications: <5000m (>3000m, derating).
- Multiple communication interfaces (including CAN, RS485, RJ45, and optical fiber, etc.) to support various communication methods.

2.2 Appearance

Figure 2.2-1 Appearance of the MV skid

The MV skid mainly consists of a Ring Main Unit (RMU), a 33kV/0.69kV transformer, four sets of 2500kW PCSs, and a container.

2.3 Electrical Structure

The electrical structure of the MV skid is shown in the following figure. Bidirectional converter topology is adopted to realize bidirectional power flow for charging and discharging.

In this structure, when the energy storage unit is charging, the MV skid operates as a rectifier, which converts the system side AC to DC, and stores the energy into the energy storage unit. When the energy storage unit is discharging, the MV skid operates as an inverter, which converts the energy released by the energy storage unit from DC to AC, and feeds it back to the external system. The PCS (Power Conversion System) AC CB (circuit breaker) is capable of realizing the grid-connected control function. The output of the converter is connected to a step-up transformer, so as to match the voltage of the parallel AC network, and to isolate the energy storage system from the external system. The RMU (Ring Main Unit) is equipped with protection devices with measurement and control function, so as to ensure the normal operation of the MV skid reliably.

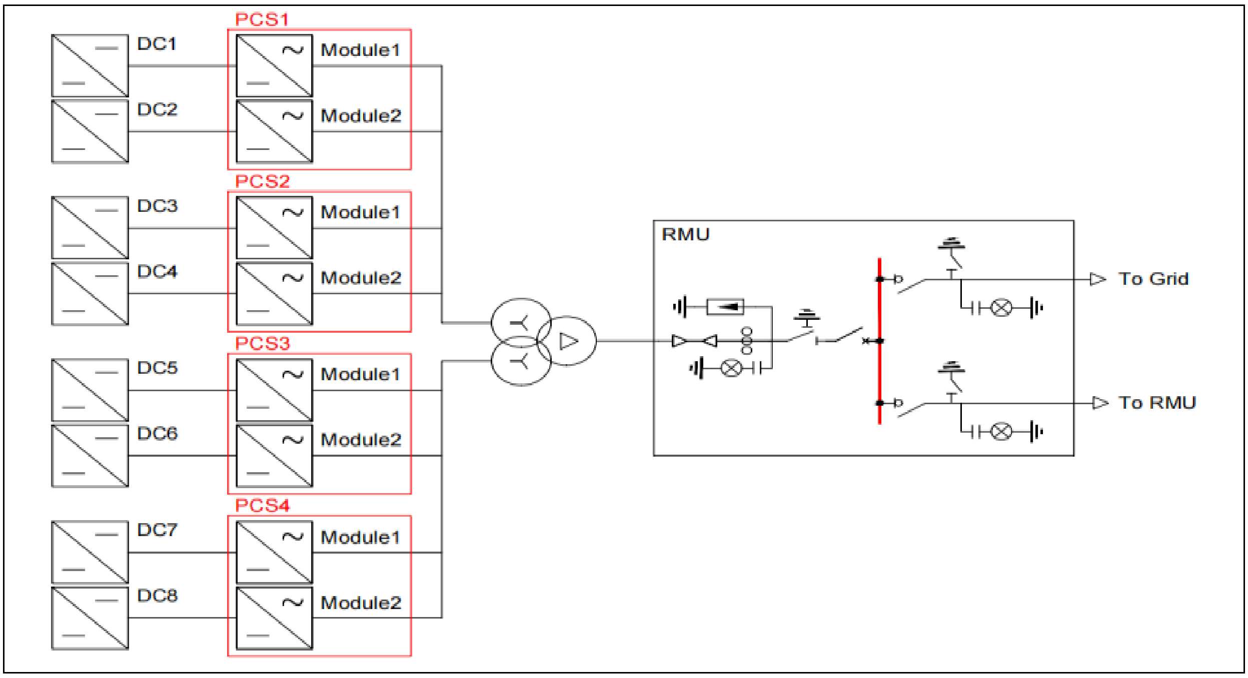

Figure 2.3-1 Main diagram of MV skid

The external 33kV bus is connected via RMU, converted to 690V AC voltage by transformer, and then connected to the AC side of the four PCSs. Meanwhile, the DC sides of the four PCSs are connected to corresponding battery stacks respectively.

2.4 Communication Topology

PCS-9567 PCS control unit (abbreviated as **"PCS-9567 control unit"** hereinafter) communicates with the BMS (Battery Management System) in real-time to obtain the status information of the batteries, and operates in charging or discharging mode according to the BMS monitoring data and control commands, achieving the purpose of energy transmission between the batteries and the AC grid. Meanwhile, the control unit can monitor the operation conditions of the whole power conversion system, ensure its reliable operation under overvoltage, overcurrent, BMS fault conditions and etc.

- **Communication with EMS**

  PCS-9567 control unit and PCS-9611S can communicate with EMS via RS-485 or Ethernet, and various protocols are supported: IEC-60780-103, IEC-61850, Modbus, and so on.

  PCS-9567 control unit supports fiber optic high-speed communication with the upper coordination controller and GOOSE protocol. It can cooperate with the upper coordination controller to form a high-speed control network and realize the fast control function.

- **Communication with BMS**

  The control unit can communicate with the BMS system, monitor real-time status of the batteries and provide batteries alarm and fault protection, ensure safe operation of batteries. It supports CAN, Modbus TCP and Modbus RTU communication modes.

By default, this product communicates with BMS in accordance with Modbus TCP protocol, which has the characteristics of fast communication speed and high stability. If adopting Modbus RTU protocol, Modbus gateway (optional) is required to realize the conversion between Modbus TCP and Modbus RTU.

The Modbus protocol gateway includes an Ethernet port and a serial port (RS-232/RS-422/RS-485). It can automatically translate between Modbus TCP (Ethernet) protocol and Modbus ASCII/RTU (serial) protocol without additional programming or work.

- **Typical communication system**

The typical communication system of the MV skid is shown in the following figure. It realizes communication with the EMS and BMS via Ethernet.

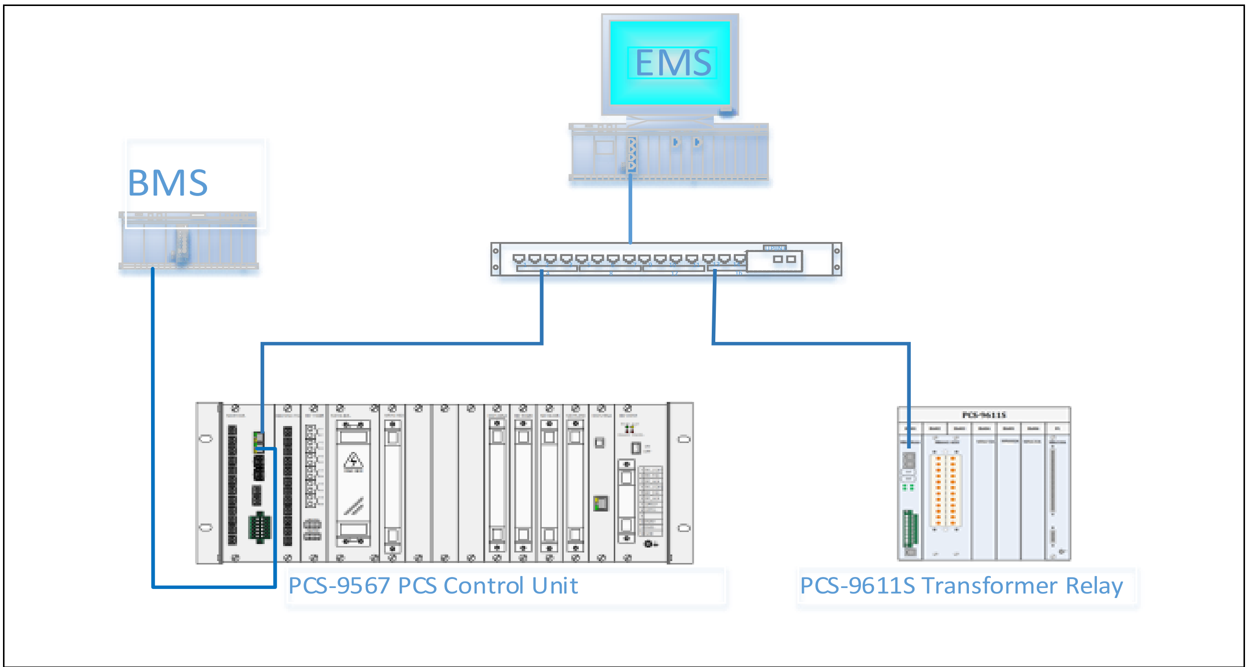

Figure 2.4-1 Typical network communication structure

2.5 Operation Modes

PCS-9567 power conversion system supports 2 operation modes: **"grid-following mode"** and **"grid-forming mode"**.

2.5.1 Grid-following Mode

In the **"grid-following mode"**, PCS-9567 control unit controls the charging and discharging of energy storage unit according to the following charging and discharging modes.

- **Constant DC voltage mode**

When this mode is enabled, the PCS will charge and discharge the energy storage unit according to the pre-configured constant DC voltage value.

- **Constant AC power mode**

When this mode is enabled, the PCS will charge and discharge the energy storage unit according to the pre-configured constant AC power value.

- **Automatic switching into constant voltage mode at the end of discharging or charging**

  When this mode is enabled, the PCS will automatically switch to the constant DC voltage mode to charge and discharge the energy storage unit at the end of discharging or charging. It is necessary to set the limits for recognizing the end of charging and discharging according to the specific battery characteristics, as well as the corresponding voltage value after switching to constant DC voltage mode.

In the **"grid-following mode"**, the PCS will charge and discharge the energy storage unit according to the pre-configured internal control parameters. The charging and discharging modes can be switched freely. The relevant configuration (including the operation mode and parameters) can be modified on the local HMI or remote HMI.

The above operation modes need to be carried out under the premise of ensuring the safety of energy storage unit, and the PCS will continuously exchange information with the BMS to confirm the safe operating range of the energy storage unit. If the current operation state exceeds the safe operating range of the energy storage unit, the PCS will automatically switch the charging/discharging current or power to the safe area provided by the BMS.

2.5.2 Grid-forming Mode

When **"grid-forming mode"** is enabled, the PCS will simulate the operation characteristic of virtual synchronous generator: change the active power by changing the power angle, and change the reactive power by changing the voltage amplitude. Its overall characteristics of voltage source can realize the natural release of energy. Meanwhile, it has certain self-discipline ability to meet the needs of parallel operation of multiple voltage sources. In this mode, the external characteristics of the energy storage converter are similar to those of the conventional synchronous unit, it has the main functions of inertia support, frequency regulation and voltage regulation, and increasing the short circuit capacity of the conventional unit. It can solve or improve the problems of low inertia and insufficient support capacity of the current power grid, and enhance the support capacity of the regional power grid. This control technology can really play a supporting role of voltage source in the power grid.

The PCS in **"grid-forming mode"** can automatically operate in **"grid-connected state"** or **"off-grid state"**.

- **"Grid-connected state" to "Off-grid state"**

  When the power grid is de-energized, or the PCC (Point of Common Coupling) switch is open, the PCS will switch to the **"grid-forming mode"** automatically or according to the grid-forming command and continue to provide the stored energy for the load.

- **"Off-grid state" to "Grid-connected state"**

  When the power grid recovers to the **"grid-connected state"**, or switch to **"grid-connected state"** by schedule, the PCS will synchronize with the power grid automatically or according to the grid-connected command, and the PCC switch will be closed.

2.5.3 Switch Operation Mode

PCS can realize online switching between **"grid-following mode"** and **"grid-forming mode"** according to actual requirements without shutdown operation.

2.6 Operation States

PCS-9567 control unit can operate in these states: **"standby state"**, **"running state"**, **"fault state"**, **"emergency stop state"** and **"shutdown state"**.

2.6.1 Standby State

When the PCS operate in **"standby state"**, if local or remote operation state switching command is received, it can switch to corresponding state quickly.

2.6.2 Running State

In **"running state"**, the PCS can operate in **"grid-connected mode"** or in **"independent inverter mode"**, and it can realize bidirectional power flow in grid-connected running.

2.6.3 Fault State

If any fault is detected, the PCS will automatically stop running and enter the **"fault state"**, and the fault information will be displayed on the LCD of the control unit for the user to check. The system continuously monitors whether the fault is eliminated or not, if so, the system will switch to the **"shutdown state"** and wait for a startup command, if not, the system will remain the **"fault state"**.

2.6.4 Emergency Stop State

When pressing the **"EMERGENCY STOP"** button, the system will enter the **"emergency stop state"**. After pressing the **"EMERGENCY STOP"** button, switch the AC circuit breaker to the **"OFF"** position. When it is confirmed that the power conversion system is allowed to run again, switch the AC circuit breaker to the **"ON"** position.

2.6.5 Shutdown State

The shutdown state means the PCS has been stopped according to the normal stop process manually.

2.7 Protection Functions of PCS-9567 Control Unit

PCS-9567 PCS control unit detects the operation state of the PCS, energy storage unit and power grid in real-time. Once a fault or abnormality is detected, the control unit will issue the corresponding alarm signal or stop command.

- **AC overvoltage protection**

  During the operation of converter, the default max permissible power grid AC voltage at the power grid interface is 110% of the rated voltage. The max permissible AC voltage can be configured by corresponding setting.

When the AC voltage is greater than corresponding setting, the converter will be stopped, and then disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **AC undervoltage protection**

During the operation of converter, the default min permissible power grid AC voltage at the power grid interface is 90% of the rated voltage. The min permissible AC voltage can be configured by corresponding setting.

When the AC voltage is lower than corresponding setting, the converter will be stopped, and then disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **AC overfrequency/underfrequency protection**

During the operation of converter, the default permissible range of the power grid frequency at the power grid interface is 45Hz~55Hz (when the rated frequency is 50Hz), or 55Hz~65Hz (when the rated frequency is 60Hz). The permissible range of the power grid voltage can be configured by corresponding setting.

When the power grid frequency exceeds corresponding setting range, the converter will be stopped. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **AC overcurrent protection**

During the operation of converter, the default max permissible AC current output is 110% of the rated current. The max permissible AC current output can be configured by corresponding setting.

When the AC current output is greater than corresponding setting, the converter will be stopped, and then disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Negative-sequence overcurrent protection**

During the operation of converter, when the AC side negative-sequence current is greater than corresponding setting, the converter will be stopped. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Negative-sequence overvoltage protection**

During the operation of converter, when the AC side negative-sequence voltage is greater than corresponding setting, the converter will be stopped. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Islanding detection**

The converter adopts dual islanding detection: active detection and passive detection. When islanding fault is detected, the energy storage unit will be disconnected from the power grid within 0.2~2s. Corresponding alarm signal will be issued and displayed on the LCD of PCS-

9567 PCS control unit.

- **DC overvoltage protection**

The default max permissible DC side voltage input is 1500V. The max permissible DC side voltage input can be configured by corresponding setting.

When the DC side voltage input is greater than corresponding setting, the converter will be stopped, the energy storage unit will be disconnected from the power grid within 0.2~1s. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **DC neutral-point voltage unbalance protection**

When the difference between the positive and negative voltages of DC side is greater than corresponding setting, the converter will be stopped instantaneously, the energy storage unit will be disconnected from the power grid within 1s. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **DC undervoltage protection**

During the operation of converter, when the DC side voltage input is lower than corresponding min permissible voltage setting, the converter will be stopped instantaneously. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Reverse connection protection of DC input power cable**

During the operation of converter, if the DC input power cable is reversely connected, the converter will be stopped instantaneously and the DC switch will be opened. Only when the connection is correct, the converter will be put into operation.

- **DC overcurrent protection**

During the operation of converter, when the DC side current is greater than corresponding max permissible setting, the converter will be stopped within 0.2~1s. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit. This setting should be configured in coordination with the charging and discharging limitation current of the energy storage unit.

- **Insulation detection**

Before startup of the converter, the insulation condition of DC side is monitored in real time. If insulation abnormity is detected, the converter can NOT be started. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Resonance protection**

During the operation of converter, the three-phase voltages on the power grid side are monitored in real-time. When the harmonic component of any phase voltage on the power grid side exceeds the limit, or the number of zero-crossing points of any phase current exceeds 250 times within 1 second, or any phase capacitor current is greater than the overcurrent protection setting, the converter will be stopped. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Driving protection**

  During the operation of converter, the state of IGBT (Insulated Gate Bipolar Transistor) module is monitored in real time. If the driving fault of IGBT is detected, the converter will be stopped instantaneously. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **VT (VoltageTransformer) supervision**

  During the operation of converter, the voltage deviation between the two sides of the AC circuit breaker is monitored in real time. If any abnormal voltage measurement is detected, the converter will be stopped instantaneously. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Auxiliary power supply protection**

  During the operation of converter, the state of PCS-9567-PWA auxiliary DC power supply is monitored in real time. If any auxiliary DC power fault is detected, the converter will be stopped instantaneously. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **Over temperature protection**

  During the operation of converter, the temperature of power modules is monitored in real time. When the temperature exceeds the upper setting, the air cooling will be started automatically and the converter will operate according to the pre-configured power range. When the temperature is greater than the ultra upper setting, the converter will be stopped, and it can resume running when the temperature recovers to normal range.

- **Communication failure protection**

  During the operation of converter, it monitors the communication state between PCS-9567 PCS control unit and BMS/EMS), and the heartbeat messages (which requires the support of EMS) in real time. When the communication is interrupted, the converter will be stopped. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **External interlock protection**

  The converter can connect with the external interlocking protection signal. When the external interlocking protection operates, the converter will be stopped. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9567 PCS control unit.

- **External emergency shutdown protection**

  External emergency stop protection signal can be connected to the converter. When the external emergency stop signal is **"1"**, the converter will be stopped. Meanwhile, the AC CB (Circuit Breaker) and DC switch will be tripped, and the corresponding alarm message will be issued.

2.8 Protection Functions of PCS-9611S

- Overcurrent protection

  During the operation of the converter, the AC output current will not exceed the overcurrent protection setting. When short circuit occurs in the power grid, and the AC output current may exceed the overcurrent protection setting. The RMU (Ring Main Unit) circuit breaker will be tripped by overcurrent protection, thus disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9611S.

- Transformer over-temperature protection

  When the transformer is running normally, the temperatures of transformer winding/oil will not exceed the over-temperature protection setting. In case of transformer fault, or the ambient temperature is too high, the temperatures of transformer winding/oil may exceed the over-temperature protection setting. The RMU circuit breaker will be tripped by over-temperature protection, thus disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9611S.

- Transformer room door open protection

  During normal operation of the system, the transformer room door and high-voltage room door are forbidden to be opened for operational safety. Once PCS-9611S detects door opening during operation, the RMU circuit breaker will be tripped by transformer room door open protection, thus disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9611S.

- Transformer Buchholz trip protection

  Oil-immersed transformers will not generate gas during normal operation. If some faults (e.g.: short circuit, discharging and etc.) occur inside the transformer, the transformer oil may generate gas to trigger Buchholz trip protection. The RMU circuit breaker will be tripped by transformer Buchholz trip protection, thus disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9611S.

- Pressure relief protection

  Pressure relief protection is a kind of safety protection. If some faults (e.g.: short circuit, discharging and etc.) occur inside the transformer, a large amount of gas will be generated to increase the pressure inside the transformer, so as to trigger the transformer pressure relief protection. The RMU circuit breaker will be tripped by pressure relief protection, thus disconnected from the power grid. Corresponding alarm signal will be issued and displayed on the LCD of PCS-9611S.

3 Man-Machine Interactions

3.1 Overview

The local HMI is located at the front of the PCS cabinet, including **“STOP”** and **“RUN”** LED indicator, **“EMERGENCY STOP”** button, **“START/STOP”** selector switch.

Figure 3.1-1 HMI layout diagram

- **LED indicators**

  The LED indicators are used to indicate the operation state of the PCS.

Table 3.1-1 LED indicators

No.	Name	Color	Description
1	STOP	Green	PCS is out of service
2	RUN	Red	PCS is running

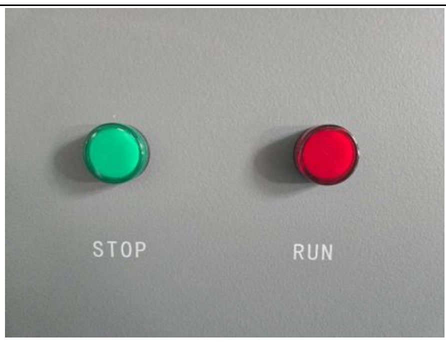

Figure 3.1-2 LED indicators

- **“EMERGENCY STOP”** button

  Press down this button, the DC switch will be disconnected, the AC circuit breaker will be tripped, and the PCS will be out of service. This button is mainly used to immediately disconnect the PCS from the power grid in case of system fault. After the fault is cleared and before the system restart, please pay attention to lift the button up. Besides, in case of normal operation of the unit, please follow the procedure for normal shutdown operation. Do NOT press the emergency shutdown button at will.

Figure 3.1-3 “EMERGENCY STOP” button

- **“START/STOP”** selector switch

  Rotate the selector switch to **“STOP”** to stop the PCS at once.

  Rotate the selector switch to **“START”** to start the PCS.

Figure 3.1-4 “STOP/START” selector switch

3.2 Power on the Device

Before powering on this product, check and confirm that the power supply is wired correctly, both AC circuit breaker and DC switch are open. Toggle the button on the power supply board of the device, and the device will be powered on.

INFORMATION

Under normal circumstances, the start-up time of the device shall not exceed 30s. If the "**STOP**" indicator on the panel of PCS is not lit on after this time, it means that there is device startup abnormity, try to power off the device and restart it. If the fault still exists, please contact the manufacturer for resolution.

3.3 Instructions for Use

After connecting the device with the PCS-PC software, the following menu can be displayed in the status bar on the left side.

3.3.1 Measurements

This menu is used to display real-time measured values of this device. Please check the detailed values of this menu before putting this device into service.

3.3.2 Status

This menu is used to display status information, binary input signals and self-supervision alarm signals. Please check the values of the signals in this menu before putting this device into service.

Table 3.3-1 Description of submenu "Status"

No.	Submenu Name	Description
1	Operation State	Display operation state information of this device.
2	Contact_Inputs	Display the binary input information of this device.
3	Superv_Events	Display the self-supervision alarm states of this device.

3.3.3 Records

Various records of this device are displayed in this menu. Please check and record these records first after device operation.

Table 3.3-2 Description of submenu "Records"

No.	Submenu Name	Description
1	Disturb Records	Display protection operation of this device.
2	Superv Events	Display supervision events of this device.
3	I/O Events	Display binary input/output events of this device.
4	Device Logs	Display device logs of this device.
5	Control Logs	Display remote control logs of this device.
6	Clear Records	Clear all records.

3.3.4 Settings

This menu is used to check and modify the parameters and settings of this device.

Table 3.3-3 Description of submenu "Settings"

No.	Submenu Name	Description
1	Comm Settings	Check or modify the communication settings.
2	System Settings	Check or modify the system parameters.
3	Logic Links	Check or modify the logic links settings.
4	Protection Settings	Check or modify the protection settings.
5	BMS Settings	Check or modify the BMS settings.

4 Routine Operations

4.1 Mode Switching

Power Conversion Systems (PCS) have multiple operating modes, which can be set according to the site requirements through the Energy Management System (EMS) or directly through the PCS-PC auxiliary software. The main operating modes include: DC voltage control mode, DC power control mode, AC power control mode, power factor control mode, and off-grid control mode.

4.2 Startup/Shutdown Operations

After the equipment is put into operation, the routine operations of PCS are generally controlled through the EMS. Functions such as startup and shutdown, remote adjustment and control, and mode switching can all be completed through the EMS.

4.2.1 Startup/Shutdown in Grid-following Mode

4.2.1.1 Startup Procedure

1) Open the grounding switch and close the load switch in the RMU incoming cabinet.
2) Open the grounding switch and close the circuit breaker in the RMU outgoing cabinet.
3) Close the1K miniature circuit breaker of the PCS internal control circuit.
4) Power on the battery side and confirm that the output terminals of the PCS DC switch are live.
5) Remotely close the DC switch of PCS.
6) Check and confirm the PCS operating mode and active power setting, and ensure that there are no alarms on the device.
7) Remotely start the PCS.

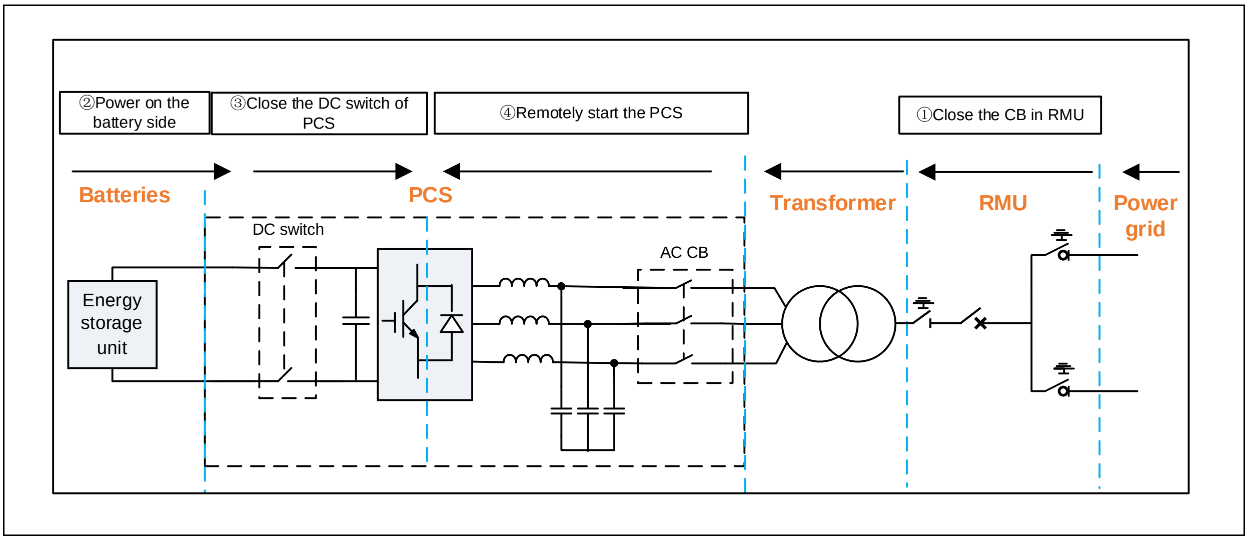

Figure 4.2-1 Startup flowchart

4.2.1.2 Shutdown Procedure

- **Routine shutdown**

1) Remotely stop the PCS

- **Long-term shutdown or maintenance shutdown**

1) Remotely stop the PCS.
2) Remotely open the PCS DC switch and monitor the PCS DC bus voltage to confirm it drops to 0.
3) Power off the battery side.
4) Turn off the power miniature circuit breaker (1K) of secondary control circuit.
5) Trip the circuit breaker and close the grounding switch in the RMU outgoing cabinet.
6) If maintenance is required, implement safety measures in accordance with regulations.

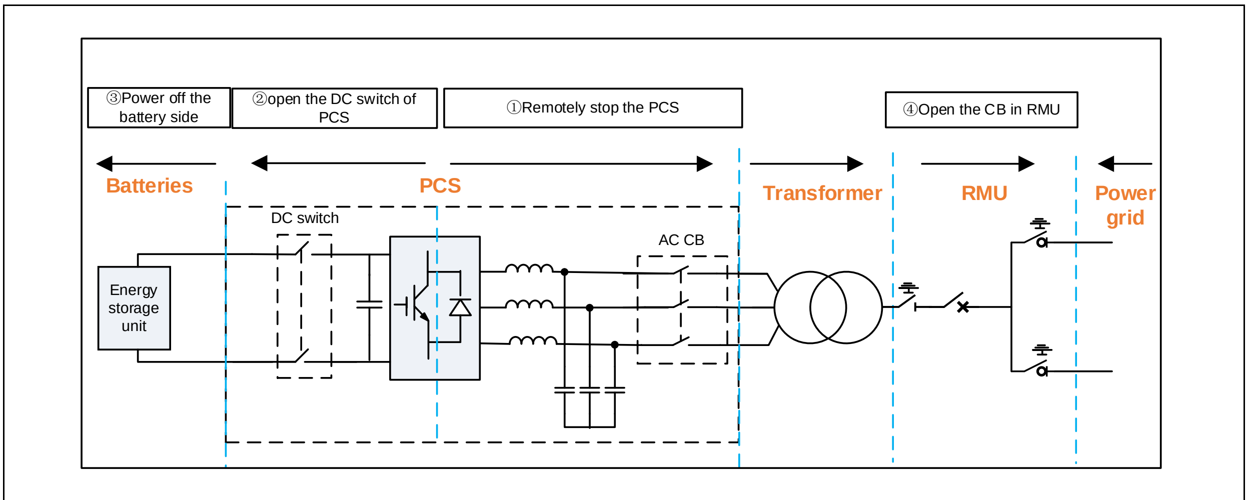

Figure 4.2-2 Shutdown flowchart

4.2.2 Startup/Shutdown in Off-grid Mode

4.2.2.1 Startup Procedure (Black Start)

1) Confirm that the high-voltage busbar connected to the MV skid is not live and that the switch at grid-connected point is in the open position.
2) Open the grounding switch and close the load switch in the RMU incoming cabinet.
3) Open the grounding switch and close the circuit breaker in the RMU outgoing cabinet.
4) Power on the battery side, ensuring the output terminals of the PCS DC switch are live.
5) Use the EMS to remotely close the DC switch.
6) Verify via the EMS that the PCS operating mode is set to off-grid mode (or grid-forming mode), no alarms are present, and the "Sig_Ready_BlackStart_Conv" signal is set to 1 (meaning Black Start Ready).
7) Remotely start the PCS (or manually operate the **"START/STOP"** selector switch on the PCS cabinet door to start the PCS).
8) After startup, observe that the AC voltage parameters are normal, and gradually increase the load, and the load power cannot be greater than the rated capacity of PCS.

⚠ CAUTION

When PCS is operating in off grid mode, it is not allowed to switch loads with large starting current, such as transformers, etc., as the large excitation inrush current can cause PCS over-current tripping. If the circuit does need a transformer, it can be put into operation before the black start, and the transformer can be slowly started in off grid mode.

4.2.2.2 Shutdown Procedure

- **Routine shutdown**

  1) Disconnect all loads from the busbar to ensure that the PCS operates in a no-load condition.
  2) Send a stop command through EMS to stop the equipment.

- **Long-term shutdown or maintenance shutdown**

  1) Disconnect all loads from the busbar to ensure the PCS operates in a no-load condition.
  2) Issue a stop command via the EMS to stop the PCS
  3) Remotely open the DC switch and monitor the PCS DC bus voltage to confirm it drops to 0.
  4) Power off the battery side.

5) Turn off the power miniature circuit breaker (1K) of secondary control circuit.

6) Open the circuit breaker and close the grounding switch in the RMU outgoing cabinet.

7) Open the load switch and close the grounding switch in the RMU incoming cabinet.

8) If maintenance is required, implement safety measures in accordance with safety regulations.

4.2.3 Emergency Stop Procedure

In case of an emergency, immediately press the **“EMERGENCY STOP”** button on the cabinet door to force the equipment to shut down.

⚠ CAUTION

（1）The **“EMERGENCY STOP”** button must ONLY be used during emergencies. For routine shutdowns, follow the standard operating procedure.

（2）When the main circuit of the equipment is live, it is forbidden to touch the AC/DC side protective barrier. Touching it may cause personal injury or equipment damage

5 Maintenance

5.1 Preventive Maintenance

5.1.1 Routine Maintenance

DANGER

DO NOT open the PCS door or transformer room door without authorization. Contact with the internal circuit may result in dangerous high voltage, causing damage or injury!

WARNING

During the PCS normal operation, **DO NOT** disconnect the AC-side circuit breaker or DC-side switch directly without authorization, as it may lead to personal injury or equipment damage!

NOTICE

DO NOT allow PCS to run under overload condition for a long time, and pay attention to the temperatures of PCS and its modules during overload operation.

NOTICE

DO NOT cut off the power supply of PCS control device during operation. If it is required to disconnect the power supply, please follow the shutdown operation procedure.

Table 5.1-1 PCS inspection item list

No.	Inspection item	Standard	Inspection cycle
1	Equipment appearance inspection	The equipment appearance is intact without damage, and the paint surface is complete without rust	Once a week is recommended
2	LED indicator inspection	The indicators work normally (The green LED indicates that the PCS is in shutdown status, the red LED indicates that the PCS is in operation status).	
3	PCS air inlet and outlet inspection	The top air duct of PCS is the air inlet, and the bottom air duct is the outlet. The inlet and outlet filters are unobstructed	

No.	Inspection item	Standard	Inspection cycle
4	Equipment operation sound	PCS operates with uniform sound and no abnormal noise	
5	Equipment operation odor	No abnormal smells during PCS operation (e.g., burning odor or smoke smell)	
6	Equipment fixation	Equipment foundation remains intact without settlement; equipment is securely fixed without displacement	
7	Surrounding environment	No weeds or debris around the equipment to avoid affecting heat dissipation	
8	Equipment grounding	Equipment grounding cables are securely connected without rust	
9	Equipment cabinet door inspection	Cabinet door locked	

Table 5.1-2 Transformer inspection item list

No.	Inspection item	Standard	Inspection cycle
1	Equipment appearance inspection	The equipment appearance is intact without damage, and the paint surface is complete without rust	Once a week is recommended
2	Oil level inspection	Check the oil level indicator to ensure that the oil level is within the normal range	
3	Oil temperature inspection	Check the oil temperature gauge to ensure that the oil temperature is within the normal range	
4	Oil color inspection	Observe the transformer oil color through the oil level meter. The transformer oil should be transparent, without turbidity, and the color should be light yellow or light brown	
5	Sound inspection	Monitor whether the transformer sound is normal during operation, and the transformer should emit a uniform "buzz" sound during normal operation. If there is an abnormal noise (such as crackling, popping), it may indicate an internal fault	
6	Equipment fixation	Equipment foundation remains intact without settlement; equipment is securely fixed without displacement	
7	Surrounding environment	No weeds or debris around the equipment to avoid affecting its normal operation	
8	Equipment grounding	Equipment grounding cables are securely connected without rust	

Table 5.1-3 RMU inspection item list

No.	Inspection item	Standard	Inspection cycle
1	Equipment appearance inspection	The equipment appearance is intact without damage, and the paint surface is complete without rust	
2	Running status check	Check whether the RMU running indicators are normal. Verify that the position of each switch (closed, open, grounded) is correct. The protection device has no alarm.	
3	Sound inspection	Monitor whether the transformer sound is normal during operation. During normal operation, the RMU should have no abnormal noise (such as discharge sound and vibration sound).	Once a week is recommended
4	Equipment fixation	Equipment foundation remains intact without settlement; equipment is securely fixed without displacement	
5	Surrounding environment	No debris around the equipment to avoid affecting its normal operation	
6	Equipment grounding	Equipment grounding cables are securely connected without rust	

5.1.2 Shutdown Maintenance

Before maintenance, please shutdown the equipment according to the normal process, and then disconnect the AC power supply, DC power supply, and circuit breaker in RMU, so that the MV skid is in maintenance state. Observe the DC bus voltage and operate after decreasing the voltage of the DC bus capacitor to a safe range through the discharge circuit.

⚠ DANGER

During maintenance, ensure the PCS cannot be accidentally powered on. It is recommended to take insulation protective measures when touching potentially live parts.

⚠ WARNING

In case of long-term shutdown, pay attention to the ventilation of the container or the room. Before entering the cable trench for maintenance, ensure proper ventilation.

NOTICE

In harsh environments with heavy dust/sand, promptly clean the air duct filters based on conditions.

INFORMATION

Record all abnormal (incident) conditions observed during operation since commissioning, and prioritize inspection of components with prior anomalies during maintenance.

Table 5.1-4 PCS maintenance item list

Maintenance item	Maintenance method	Maintenance cycle
Appearance inspection	1. Clean the dust on the main circuit equipment, such as copper bar, circuit breaker, power module, filter capacitor, stacked busbar, etc. It can use a vacuum cleaner and brush to remove dust.	The recommended maintenance cycle is twice a year, and once every two months for power stations with high summer temperatures or continuous heavy loads.
	2. Clean the dust on the secondary equipment, such as the PCS-9567 control device, auxiliary power supply, Hall sensor, VT, CT, contactor, voltage divider resistor box, insulation monitoring device, etc.	
	3. Clean the dust inside the air duct filter to ensure good ventilation. Remove the air duct filter and clean it thoroughly with a vacuum cleaner.	
	4. Check the PCS cabinet door to ensure smooth operation of the door lock switch without affecting normal use.	
	5. Check whether the surrounding environment is unfavorable for operation and perform weed removal in the vicinity.	
	6. Check whether the PCS cabinet is intact, whether there are defects such as overheating, contact discoloration, deformation, or vibration.	
	7. Verify that cable entry holes are properly sealed and reinforce sealing as needed to prevent small animals from entering.	
	8. Check the PCS air inlets/outlets to ensure no obstructions are present.	
	9. Check for no condensation or water ingress inside the equipment.	
Primary circuit inspection	1. Check whether the cable insulation layer has signs of deterioration, cracking, peeling or overheating.	
	2. Check whether the cable connection is loose.	
	3. Check whether the connection between cables or between cable and copper bar has discolored. If it turns yellow or black, it may be caused by overheating.	
	4. Check whether the components (such as copper strips, screws, and bolts) are loose, damaged, discolored, deformed, or have discharge marks.	

Maintenance item	Maintenance method	Maintenance cycle
	5. Ensure that all busbars, cables, and connections are clean and dry.	
	6. Check the grounding resistance load of the equipment according to local standards.	
Secondary and communication circuit inspection	1. Check that the wiring harnesses of the secondary transformer, contactor, miniature circuit breaker, sampling device, etc. inside the PCS are securely connected.	
	2. Check that the pulse synchronous optical fiber between PCS is not bent or damaged.	
	3. Check that the communication network cables between PCS and EMS/PMS are securely connected and undamaged.	

Table 5.1-5 Transform maintenance item list

Maintenance item	Maintenance method	Maintenance cycle
Appearance inspection	1. Clean up dust and debris. Use a soft brush or vacuum cleaner to remove dust and debris from the surface of the transformer casing.	Once a year
	2. Clean oil stains. Use a cloth dipped in cleaning agent to wipe off the oil stains on the surface of the transformer casing. For stubborn oil stains, use a specialized cleaning agent for removal.	
	3. Clean bushings. Use a clean cloth to clean the high-voltage and low-voltage bushings, ensuring they are free of dust and oil stains. Inspect the bushings for cracks or discharge marks, and promptly address any abnormalities.	
	4. Check the surrounding environment. Ensure there is no debris nearby and clear weeds and grass in the surrounding area.	
Primary circuit connections inspection	1. Inspection of copper bars and cable fastening. Check the copper bars and cable connections on the high and low voltage sides to ensure they are securely connected and have not discolored.	
Inspection of important components	1. Check that the gas relay operates normally.	
	2. Check that the pressure relief valve operates normally.	
	3. Check the thermometer and ensure the temperature display is normal.	
	4. Check the oil level indicator, and ensure the oil level display is normal.	
Oil quality inspection	Transformer oil samples should be sent to local institutions for testing, and should meet the local oil quality standards.	

Maintenance item	Maintenance method	Maintenance cycle
Insulation resistance test	Use an insulation resistance tester (such as a megohmmeter) to measure the insulation resistance of high voltage windings, low voltage windings, and windings to ground.	
Voltage withstand test	Perform power frequency withstand voltage test or induction withstand voltage test on high voltage winding and low voltage winding respectively	

Table 5.1-6 RMU maintenance item list

Maintenance item	Maintenance method	Maintenance cycle
Appearance inspection	1. Check whether the RMU enclosure is intact, free of deformation, rust or damage.	Once a year
	2. Verify the cabinet doors, seals, observation windows, etc. are intact and not damaged.	
	3. Ensure the grounding device is secure and reliable	
	4. Check the surrounding environment and ensure that there are no debris in the hallway	
Primary circuit connections inspection	1. Inspect the copper bars and cable connections to ensure that they are securely connected and have not discolored.	
Status check	1. Check whether the operation indicators of RMU are functioning normally.	
	2. Confirm the positions of all switches (closed, open, grounded) are correct.	
	3. Ensure no alarms are present in protection and control devices.	
	4. Verify that the local and remote opening and closing operations of the switchgear are functioning normally.	
	5. Confirm that the switch energy storage unit is operating normally.	
Electrical protection test	1. Perform overcurrent protection testing to confirm reliable switch operation	
Mechanical alarm and protection test	1. Transformer high-temperature alarm	
	2. Transformer overtemperature trip	
	3. Transformer low oil level alarm	
	4. Transformer high oil level alarm	
	5. Transformer pressure relief trip	
	6. Transformer light gas alarm	

Maintenance item	Maintenance method	Maintenance cycle
	7. Transformer heavy gas trip	
	8. Switchgear blockage	
	9. Switchgear alarm	
	10. Low-voltage room smoke detector activation	
	11. Low-voltage room temperature sensor activation	
	12. High-voltage room smoke detector activation	
	13. High-voltage room temperature sensor activation	

5.2 Fault Maintenance

5.2.1 Fault Information and Troubleshooting

Table 5.2-1 PCS fault information and troubleshooting

No.	Fault item	Fault cause	Troubleshooting	Remarks
1	76.Op	DC current is higher than the corresponding setting	Check whether the PCS power module appearance is intact. If any components are damaged, please contact the manufacturer for after-sales service.	If the PCS is in good condition, it can run again
2	59DC.Op	DC voltage is higher than the corresponding setting	Check the DC side voltage of batteries and measure the DC busbar voltage. If the voltage is actually higher than the corresponding setting, reduce the number of batteries in series and reduce the DC side voltage of batteries; If the measured busbar voltage is normal, check the auxiliary contacts of contactor, divider resistor box, please contact the manufacturer for after-sales service if any components are damaged	If the PCS is in good condition, it can run again
3	27DC.Op	DC voltage is lower than the corresponding setting	Check whether the DC switch is closed. Check the DC side voltage of batteries and measure the DC busbar voltage, if the voltage is lower than the corresponding setting, increase the number of batteries in series. If the measured busbar voltage is normal, check the auxiliary contacts of	The fault will automatically clear once the DC voltage is restored to normal

No.	Fault item	Fault cause	Troubleshooting	Remarks
			contactor and the divider resistor box, please contact the manufacturer for after-sales service if any component is damaged	
4	RevDC-Polar.Op	The polarities of DC incoming line are reversed	Check DC incoming line	If the PCS is in good condition, it can run again
5	51P.Op	Phase current is higher than the corresponding setting	Check whether the PCS power module appearance is intact. If any components are damaged, please contact the manufacturer for after-sales service.	If the PCS is in good condition, it can run again
6	50Q.Op	Negative-sequence current is higher than the corresponding setting	Check whether the PCS power module appearance is intact. If any components are damaged, please contact the manufacturer for after-sales service.	If the PCS is in good condition, it can run again
7	59P.Op	Phase voltage of the power grid is higher than the corresponding setting	Check the location of AC circuit breaker and check for power grid faults. Check whether the PCS power module appearance is intact. If any components are damaged, please contact the manufacturer for after-sales service.	If the PCS is in good condition, it can run again
8	27P.Op	Phase voltage of the power grid is lower than the corresponding setting	Check the location of AC circuit breaker and check for power grid faults.	The fault will automatically clear once the DC voltage is restored to normal
9	59Q.Op	Negative-sequence voltage is higher than the corresponding setting	Check for power grid faults.	If the PCS is in good condition, it can run again
10	81O.Op	System frequency is higher than the corresponding setting	Check for power grid faults.	If the PCS is in good condition, it can run again
11	81U.Op	System frequency is lower than the corresponding setting	Check for power grid faults.	If the PCS is in good condition, it can run again
12	26.Op	The PCS temperature is out of limit	Check whether the fan is working normally and whether the room temperature is within the normal range. If the fan or other components are damaged, please contact	If the PCS is in good condition, it can run again

No.	Fault item	Fault cause	Troubleshooting	Remarks
			the manufacturer for after- sales service	
13	LowTemp.Op	The PCS temperature is lower than the corresponding setting	Check the ambient temperature	If the PCS is in good condition, it can run again
14	LVRT-Prot.Op	Low voltage ride-through failure or the low voltage ride-through time has expired	Check for power grid faults.	If the PCS is in good condition, it can run again
15	Op_Drive	IGBT failure	**Do not attempt to restart the PCS in case of this fault!** Check if the appearance of PCS power module is intact and contact the manufacturer's after-sales service.	
16	Op_Drive_A	IGBT failure in phase-A bridge arm	**Do not attempt to restart the PCS in case of this fault!** Check if the appearance of PCS power module is intact and contact the manufacturer's after-sales service.	
17	Op_Drive_B	IGBT failure in phase-B bridge arm	**Do not attempt to restart the PCS in case of this fault!** Check if the appearance of PCS power module is intact and contact the manufacturer's after-sales service.	
18	Op_Drive_C	IGBT failure in phase-C bridge arm	**Do not attempt to restart the PCS in case of this fault!** Check if the appearance of PCS power module is intact and contact the manufacturer's after-sales service.	
19	VTS.Op	Inconsistent three-phase voltage sampling at both ends of the AC circuit breaker	Check whether the PCS power module appearance is intact. If any components are damaged, please contact the manufacturer for after-sales service.	
20	AuxPwrProt.Op	Auxiliary power supply failure	Check if the auxiliary power indicator is functioning properly. If the indicator is abnormal, contact the manufacturer's after-sales service.	
21	Alm_Contactor	The contactor fails to close properly, or the	Replace the contactor or auxiliary contacts	

No.	Fault item	Fault cause	Troubleshooting	Remarks
		auxiliary contacts have poor contact		
22	26.Alm	The thermistor PT100 is abnormal, or the fan fails to rotate, or the air inlet volume decreases	Replace the power module or replace the fan, or clean the filter screen and filter cotton at the air inlets.	
23	Alm_FanFail	The power supply of fan is missing, the fan contactor cannot be closed, or the fan itself is abnormal	Measure the power supply circuit of the fan, replace the fan contactor or the fan	
24	Fail_Device	CPU plug-in module NR1101E is abnormal	Contact the manufacturer to replace the NR1101E plug-in module	

Table 5.2-2 PCS-9611S fault information and troubleshooting

No.	Fault item	Fault cause	Troubleshooting	Remarks
1	50/51P.Op	AC current is higher than the corresponding setting	1. Check the PCS alarm information to determine whether the overcurrent action is caused by a PCS fault. 2. If there is no fault in PCS, it is necessary to check whether the transformer is abnormal.	After troubleshooting, the circuit breaker can be reclosed.
2	Alm_OverTemp	Transformer oil temperature is too high, exceeding the alarm threshold	1. It is recommended to shut down the MV skid first. 2. Check if there are weeds or debris around the transformer that affect heat dissipation. 3. Check if the transformer has been operating in an overloaded state for a long time.	After troubleshooting, the circuit breaker can be reclosed.
3	Op_OverTemp	Transformer oil temperature is too high, exceeding the corresponding protection setting	1. Check if there are weeds or debris around the transformer that affect heat dissipation. 2. Check if the transformer has been operating in an overloaded state for a long time.	After troubleshooting, the circuit breaker can be reclosed.
4	Alm_OilLev_L	Transformer oil level is too low, below the alarm threshold	1. Check if there is any oil leakage in the transformer. 2. Check if the oil level gauge is	After troubleshooting, the circuit

No.	Fault item	Fault cause	Troubleshooting	Remarks
			normal.	breaker can be reclosed.
5	Alm_OilLev_H	Transformer oil level is too high, above the alarm threshold	1. Check if the oil level gauge is normal.	After troubleshooting, the circuit breaker can be reclosed.
6	Alm_Gas	Gas content inside the oil transformer exceeds the alarm threshold	1. Shut down the MV skid. 2. Check if the gas relay is functioning properly. 3. Oil sample inspection to determine if the oil sample has deteriorated.	After troubleshooting, the circuit breaker can be reclosed.
7	Op_Gas	Gas content inside the oil transformer exceeds the operation threshold	1. Check if the gas relay is functioning properly. 2. Oil sample inspection to determine if the oil sample has deteriorated.	After troubleshooting, the circuit breaker can be reclosed.
8	Alm_TempDetect	Temperature in the HV/LV room is abnormally high	1. Check if the ambient temperature in the HV/LV room is too high.	After troubleshooting, the circuit breaker can be reclosed.
9	Alm_SmokeDetect	Smoke detected in the HV/LV room	1. Check if there is smoke in the HV/LV room.	After troubleshooting, the circuit breaker can be reclosed.

5.2.2 Component Replacement

5.2.2.1 PCS Module Communication Optical Fiber Fault

When a communication fiber fault occurs in the PCS module, the drive protection operation will be triggered, and the fiber needs to be inspected and replaced.

Before maintenance, please be sure to turn the PCS into the maintenance state, confirm that the AC circuit breaker and DC switch of the PCS are disconnected, and the battery system is powered off. If a LV cabinet is configured on the LV side of the transformer, open the circuit breaker of the LV cabinet, then the PCS AC-side is disconnected from the transformer, and the PCS can be maintained. If the LV cabinet is not configured, the transformer needs to be powered off, that is, the RMU, the combined load switch or vacuum circuit breaker on the HV side of the transformer should be turned into the maintenance state.

1) By checking the alarm information from EMS and analyzing the waveform files during the protection operation, locate which phase module's drive protection has been triggered. Subsequently, conduct an on-site inspection of the communication optical fibers between the identified phase's PCS module and the device. A common and simple inspection method involves disconnecting the optical fibers at both ends and using a light source to check for any significant light attenuation. If light attenuation is evident, the optical fiber is faulty.

2) Open the rear sealing panel of the cabinet, which will expose the three-phase modules located on the middle left side. From right to left, they are Phase A, Phase B, and Phase C, respectively. These modules communicate with the device via optical fibers, all of which are labeled with digital tags (1-21).

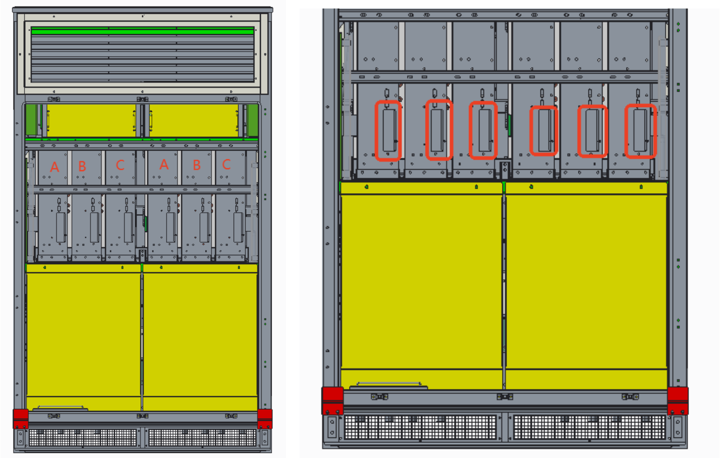

Figure 5.2-1 Optical fiber location schematic diagram

Table 5.2-3 Optical fiber numbering table

MD1-C	MD1-B	MD1-A
18	12	6
17	11	5
21	20	19
16	10	4
15	9	3
14	8	2
13	7	1

3) Open the front cabinet door, there are the B01 and B03 board used for PCS-9567 device to communicate with the module. The digital label number of the optical fiber on the board is consistent with the digital number of the optical fiber on the side of the module, indicating that it is the same optical fiber.

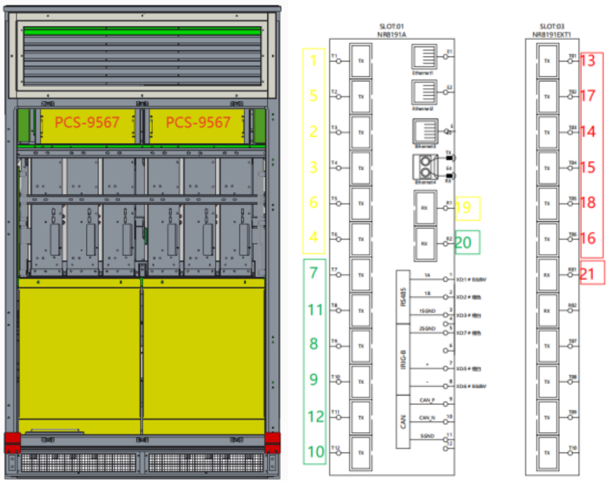

Figure 5.2-2 Optical fiber installation location diagram

4) After confirming the faulty optical fiber, replace it with a new one and ensure that the digital numbers at two ends of the optical fiber match. Before replacement, pay attention to wear ESD gloves.

5) Take proper precautions during replacement to prevent damage to other optical fibers. Lay the new optical fibers along the original path and secure them with cable ties, but do not tie them too tightly to avoid damage.

5.2.2.2 Replacement of PCS Module

When an IGBT fault occurs in the PCS module, a drive protection operation will be reported, and there may be obvious fault phenomena inside the module, requiring the whole module to be replaced.

Before maintenance, please be sure to turn the PCS into the maintenance state, confirm that the AC circuit breaker and DC switch of the PCS are disconnected, and the battery system is powered off. If a LV cabinet is configured on the LV side of the transformer, open the circuit breaker of the LV cabinet, then the PCS AC-side is disconnected from the transformer, and the PCS can be maintained. If the LV cabinet is not configured, the transformer needs to be powered off, that is, the RMU, the combined load switch or vacuum circuit breaker on the HV side of the transformer should be turned into the maintenance state.

1) By checking the alarm information from EMS and analyzing the waveform files during the protection operation, locate which phase module's drive protection has been triggered. Conduct an on-site inspection to check if there are obvious faults in the corresponding PCS module, and replace the faulty module.

2) Open the front cabinet door, and you can see the left and right three-phase modules. Remove the 15V power supply terminal and fiber optic cable of the module, and protect the wiring harness. Adjust the position of the wiring harness so that it will not come into contact with the wiring harness during the replacement process.

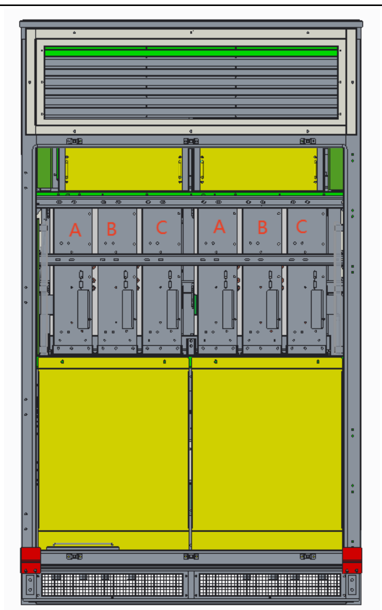

Figure 5.2-3 Power module location schematic diagram

3) Open the front cabinet door, first remove the left and right sealing plates on the AC side, then remove the module support crossbeam 1 as a whole, remove the fan connection line, then remove the module support crossbeams 2 and 3, and finally remove the air duct panel with handle. At this point, it can be seen that the copper busbar on the AC side of the module is connected to the reactor through a flexible copper busbar. Removing the flexible copper busbar between the module and the reactor completes the removal of the AC side of the module.

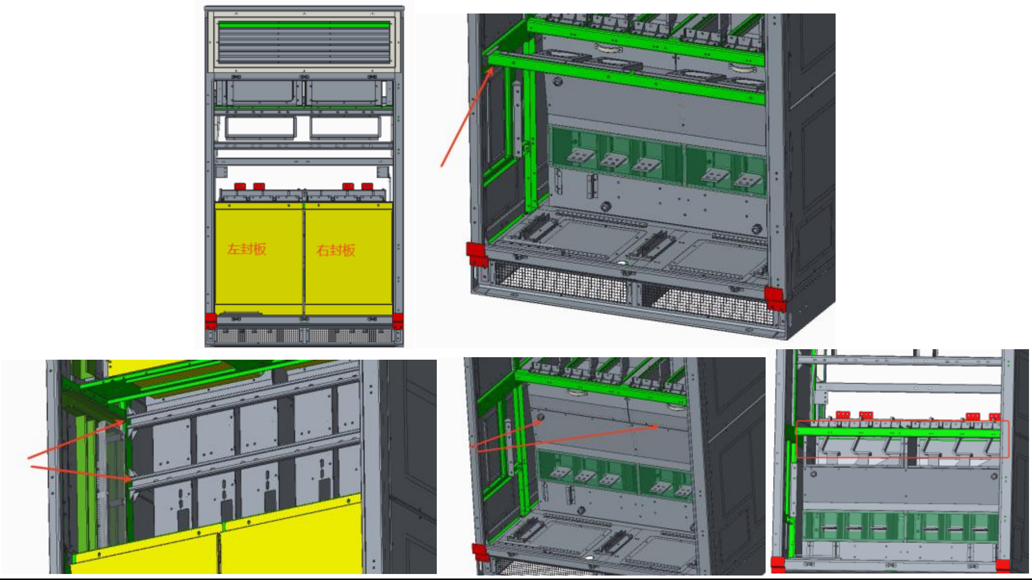

Figure 5.2-4 Power module disassembly diagram 1

4) Open the rear cabinet door, first remove the crossbeam of the capacitor module, then remove the front sheet metal of the capacitor module. At this point, it can be seen that the DC side busbar of each module is connected to the capacitor busbar through 9 screws. Remove the corresponding screws of the module that needs to be replaced, and the DC side of the module

is completed.

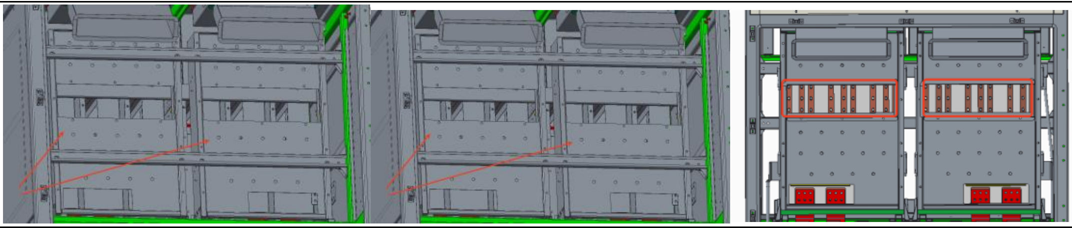

Figure 5.2-5 Power module disassembly diagram 2

5) Now the module can be removed and installed. Pull the module horizontally out from the front of the cabinet. Be careful not to touch the wiring harness during the extraction process. When removing, installing, or transporting the module, use the metal handle and housing of the module as the points of force application. Remember not to apply force to the busbar.

6) After replacing the module, reinstall the removed screws, optical fibers, and 15V power supply wiring harness, and reinstall the removed structural sheet metal sealing plate. At this point, the module replacement is complete.

5.2.2.3 Fault of PCS DC Switch

When the PCS DC switch remote control is abnormal in opening and closing, the opening and closing position is not in place, and the electric operating mechanism is removed to manually operate the DC switch, but the switch still cannot be opened and closed, it indicates that the DC switch is faulty and needs to be replaced.

Before maintenance, please be sure to turn the PCS into the maintenance state, confirm that the AC circuit breaker and DC switch of the PCS are disconnected, and the battery system is powered off. If a LV cabinet is configured on the LV side of the transformer, open the circuit breaker of the LV cabinet, then the PCS AC-side is disconnected from the transformer, and the PCS can be maintained. If the LV cabinet is not configured, the transformer needs to be powered off, that is, the RMU, the combined load switch or vacuum circuit breaker on the HV side of the transformer should be turned into the maintenance state.

1) Open the rear cabinet door and remove the upper and lower copper bars of the switch.

Figure 5.2-6 Schematic diagram 1 for replacement of DC switch operating mechanism

2) Remove the fastening screws between the DC switch and the fixed sheet metal.

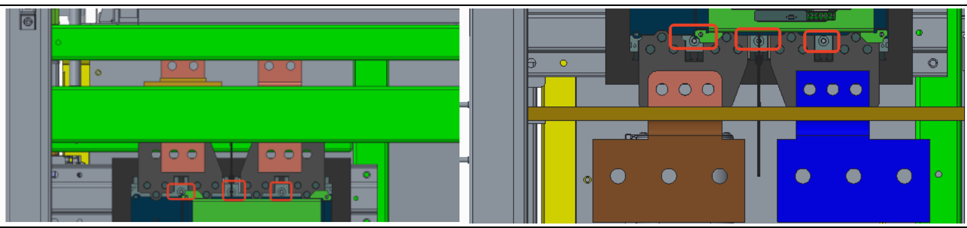

Figure 5.2-7 Schematic diagram 2 for replacement of DC switch operating mechanism

3) Install the new DC switch, restore the fixing screws .

5.2.2.4 Fault of PCS DC Hall Current Sensor

When a fault occurs in the PCS DC Hall current sensor, abnormal DC current sampling will result, and the Hall current sensor needs to be replaced.

Before maintenance, please be sure to turn the PCS into the maintenance state, confirm that the AC circuit breaker and DC switch of the PCS are disconnected, and the battery system is powered off. If a LV cabinet is configured on the LV side of the transformer, open the circuit breaker of the LV cabinet, then the PCS AC-side is disconnected from the transformer, and the PCS can be maintained. If the LV cabinet is not configured, the transformer needs to be powered off, that is, the RMU, the combined load switch or vacuum circuit breaker on the HV side of the transformer should be turned into the maintenance state.

1) Open the rear cabinet door and see the positive pole DC Hall of the DC switch. Remove the copper bar fastening at the bottom of the capacitor module and the copper bar fastening at the top of the DC switch.

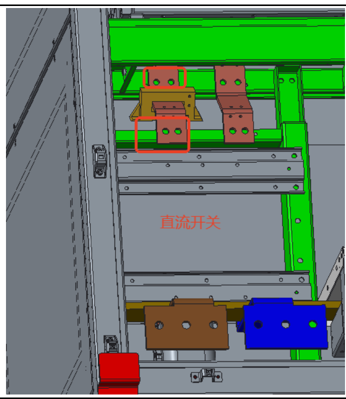

Figure 5.2-8 DC Hall sensor replacement schematic

2) Remove the secondary cables to the DC hall sensor body, and make marks before removal to ensure correct rewiring later.

3) Remove the screws on both ends of the Hall copper busbar so that the DC Hall sensor can be taken out. Take out the DC Hall sensor after removing the fixing screws.

4) Install a new Hall, restore the copper busbar screws and Hall secondary wiring, reinstall the removed copper busbar, and the DC Hall replacement is now complete.

5.2.2.5 Fault of Hall current sensor in PCS AC filter circuit

When a fault occurs in the PCS AC filter circuit Hall current sensor, abnormal capacitor current sampling will result, and the Hall current sensor needs to be replaced.

Before maintenance, please be sure to turn the PCS into the maintenance state, confirm that the AC circuit breaker and DC switch of the PCS are disconnected, and the battery system is powered off. If a LV cabinet is configured on the LV side of the transformer, open the circuit breaker of the LV cabinet, then the PCS AC-side is disconnected from the transformer, and the PCS can be maintained. If the LV cabinet is not configured, the transformer needs to be powered off, that is, the RMU, the combined load switch or vacuum circuit breaker on the HV side of the transformer should be turned into the maintenance state.

1) Open the rear cabinet door and see the Hall sensors on both sides of the AC filter capacitor.

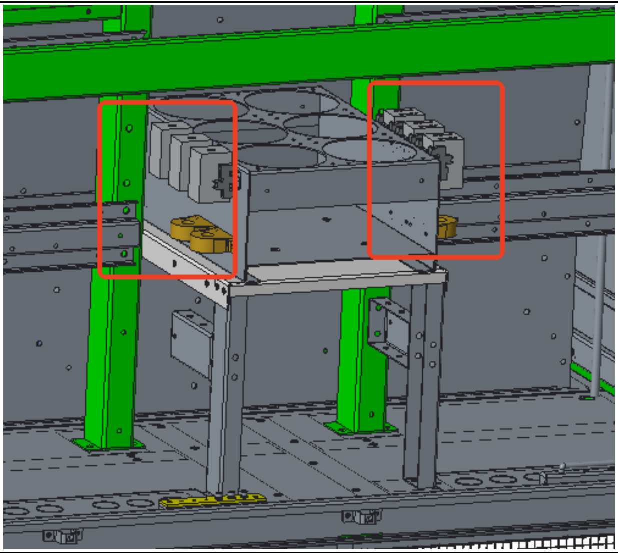

Figure 5.2-9 AC Hall sensor replacement schematic 1

2) Remove the secondary wiring of the Hall sensor body, and make marks before removal to ensure correct reconnection later.

3) Remove the cable from the terminal so that the Hall current sensor can be taken out. Take out the Hall current sensor after removing the fixing screws.

4) Install a new Hall sensor, restore the hub cable and Hall secondary wiring, reinstall and remove the Hall sensor, and now the replacement of the AC filter circuit Hall sensor is complete.

5.2.2.6 PCS Module Over-temperature

When the PCS module overheats, it may be due to a clogged filter screen caused by prolonged operation, and dust removal is required.

Before maintenance, please be sure to turn the PCS into the maintenance state, confirm that the AC circuit breaker and DC switch of the PCS are disconnected, and the battery system is powered off. If a LV cabinet is configured on the LV side of the transformer, open the circuit breaker of the LV cabinet, then the PCS AC-side is disconnected from the transformer, and the PCS can be maintained. If the LV cabinet is not configured, the transformer needs to be powered off, that is, the RMU, the combined load switch or vacuum circuit breaker on the HV side of the transformer should be turned into the maintenance state.

⚠ CAUTION

DO NOT touch the optical fiber interface of the module, do not pull the optical fiber, pay attention to the protection of the optical fiber head and the optical fiber port of the board, do not touch the surface of the driver board and the photoelectric conversion board, to avoid damaging the board.

1) Clean the outer filter screen of the air hood directly with a vacuum cleaner. If the dust is difficult to clean, remove the outer filter screen of the air hood and clean it with water.

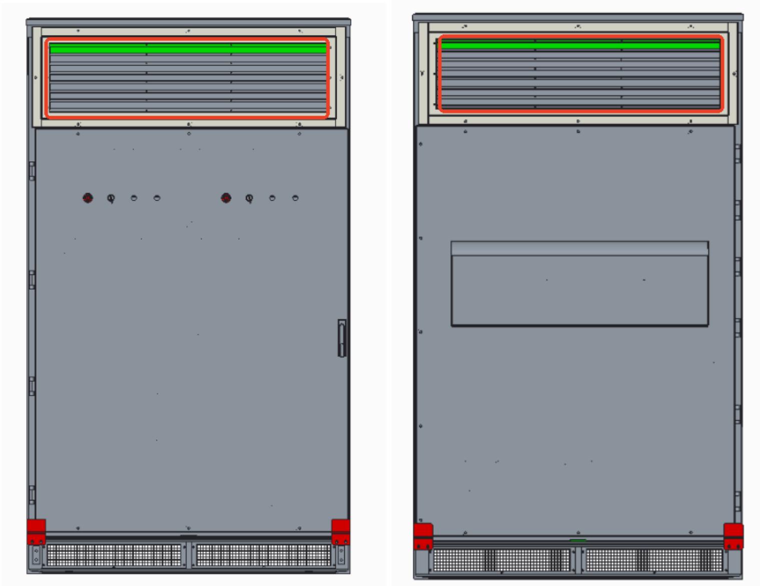

Figure 5.2-10 Air duct filter cleaning step 1

2) Clean the air outlets of the front cabinet door with a vacuum cleaner.

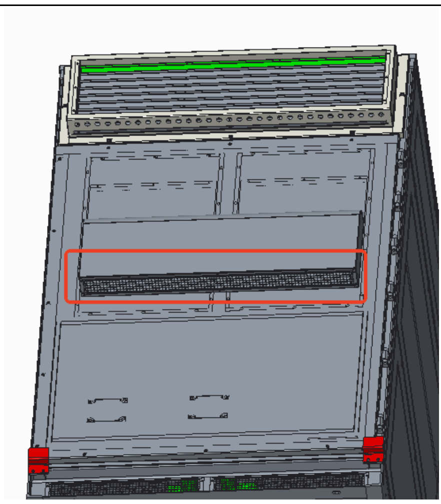

Figure 5.2-11 Air duct filter cleaning step 2

3) Check the air outlets at the bottom of the cabinet, clean the filter screens, and avoid storing items around the area to prevent obstruction of heat dissipation.

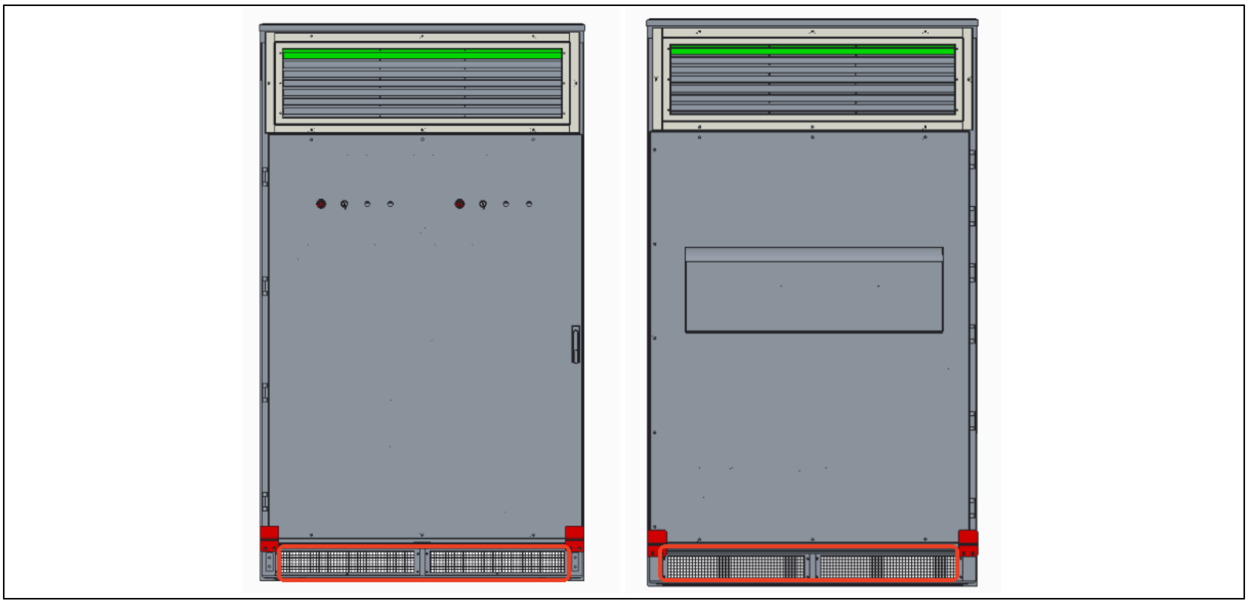

Figure 5.2-12 Air duct filter cleaning step 3

4) Open the front cabinet door and the rear sealing plate of the cabinet, and use a brush and vacuum cleaner to clean the copper bars, AC circuit breaker, DC switch, contactor, voltage dividing resistance box, module surface, resistor surface, and device surface, etc. During cleaning, avoid touching optical fibers and exposed boards to prevent damage.

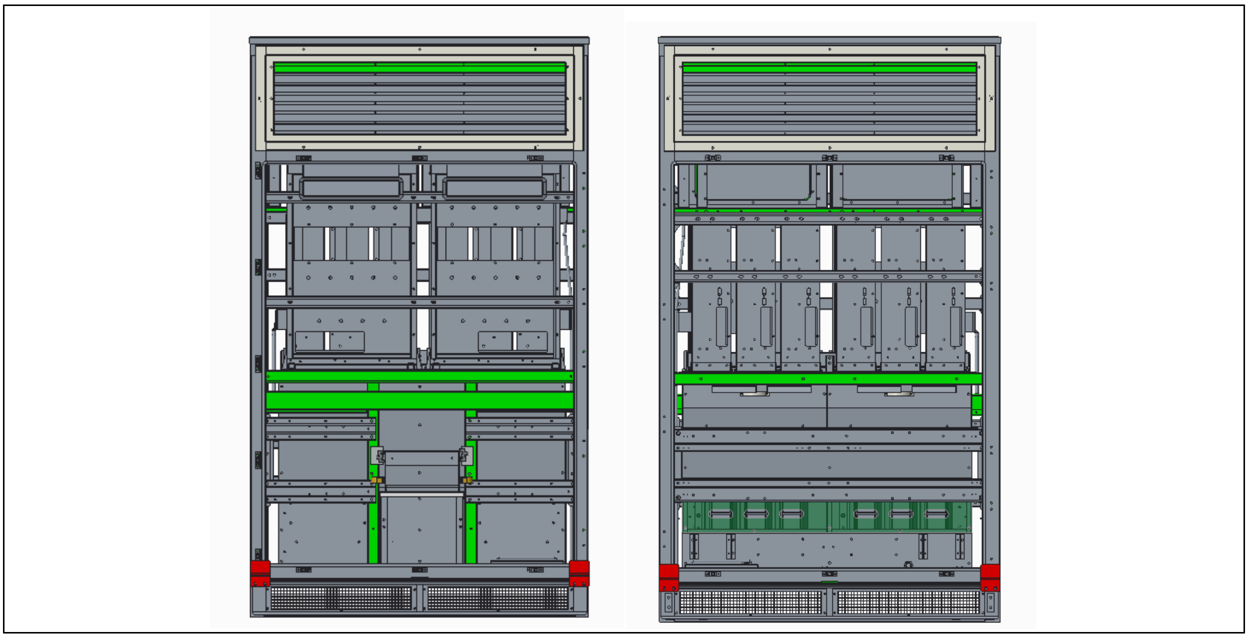

Figure 5.2-13 Equipment internal cleaning step 1

5) Clean the filter screen at the bottom of the reactor, remove the side baffle of the reactor, and clean it with a vacuum cleaner and brush.

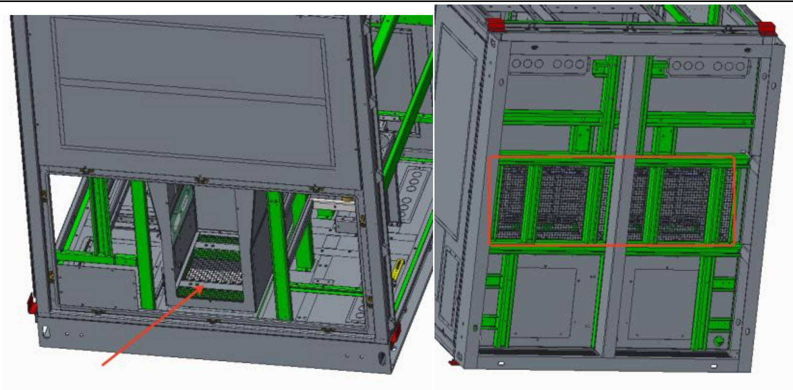

Figure 5.2-14 Equipment internal cleaning step 2

6) After the cleaning is completed, confirm that the equipment is restored to its original state, and account for all tools to avoid leaving any tools or other debris inside the cabinet.

5.2.2.7 Maintenance of Painted Surfaces

If the paint on the surface of the equipment is damaged or there are visible rust marks, it should be treated as soon as possible. If necessary, stop the equipment first and then process it. The tools required are: polishing disc, sander, flexible grinding sponge, degreasing spray, rolling brush rod, rags, hair dryer, disposable gloves, protective glasses, face mask, brush, epoxy primer, polyurethane.

1) If there are obvious rust marks on the damaged area, use sandpaper or a polishing disc to remove the rust first.

2) Clean the damaged area with a rag or hair dryer and, if necessary, use putty to flatten the surface.

3) If there are obvious rust marks on the damaged area, apply epoxy primer to the damaged area and wait for at least 24 hours for it to dry completely.

4) Apply the first coat of paint to the damaged area and wait at least 12h for it to dry completely.

5) Apply the second coat of paint to the damaged area.

6) Apply the final coat (e.g. polyurethane).

Galvanized surface maintenance:

1) Use sandpaper to grind and remove rust.

2) Clean the damaged area.

3) The damaged areas can be coated with zinc-rich paint or zinc spray.

Appendix A Acronyms and Abbreviations

AC	Alternating Current
BCU	Bay Control Unit
BI	Binary Input
BMS	Battery Management System
BO	Binary Output
CAN	Controller Area Network
CB	Circuit Breaker
CT	Current Transformer
DC	Direct Current
EMI	Electromagnetic Interference
EMS	Energy Management System
GFM	Gird-Forming Mode
HMI	Human Machine Interface
HV	High Voltage
HVS	High-Voltage Side
IEC	International Electrotechnical Commission
IGBT	Insulated Gate Bipolar Transistor
IMD	Insulation Monitoring Device
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LV	Low Voltage
LVS	Low-Voltage Side
MCB	Miniature Circuit Breaker
MV	Medium Voltage

PCB	Printed Circuit Board
PCS	Power Conversion System
PMS	Power Management System
PWM	Pulse-Width Modulation
RMU	Ring Main Unit
SOC	State of Charge
SOH	State of Health
SWG	Switchgear
THD	Total Harmonic Distortion
UPS	Uninterruptible Power Supply
VT	Voltage Transformer